



Immanuel Simon

BIOMEDICAL ENGINEERING

CONTACT

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EDUCATION

University of Waterloo
B.A Sc, Biomedical Engineering
2023 – 2028 · 3rd Year

RESEARCH

Spasticity Analysis — UW BME

Undergraduate Research Assistant processing EMG & joint-angle data to study velocity-dependent resistance during passive hand movement; MATLAB signal-processing pipeline (filtering, RMS/envelope, onset detection, regression).

TECHNICAL TOOLKIT

Programming

Python C++ Java MATLAB JavaScript SQL

Biomedical / Electrical

EMG Signal Processing Analog Circuits Instrumentation

Design & Embedded

SolidWorks Fusion 360 Arduino 3D Printing

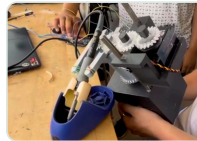
AI / Data

Machine Learning Data Analysis Git

Featured Work — Medical Devices · Biomedical Signals · Software & AI

Portfolio 2026

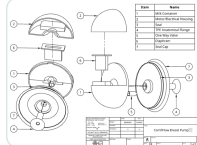
FEATURED PROJECTS



01 Solder Man — EMG-Controlled Prosthetic

EMG · Arduino · Analog Circuits · Mechanical Design

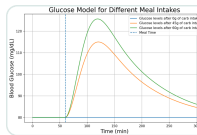
Low-cost, task-specific prosthetic that manipulates a soldering iron. Custom EMG acquisition circuit (amplification, filtering, rectification) driving a bimodal servo mechanism for grip + wrist rotation, with firmware mapping muscle intensity to motor action.



02 ComfiFlow — Wearable Breast Pump

Medical Device · CAD · Regulatory · V&V

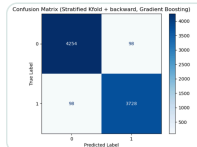
Wearable breast pump designed for comfort, hygiene, and reliability, featuring a TPE anatomical flange and closed system. Included a risk assessment, verification & validation plan, and 510(k) regulatory-pathway considerations.



03 Glucose-Insulin Simulator

Physiological Modelling · MATLAB/Python · Web

Interactive model extending a Bergman-style three-state glucose-insulin system with a new state and meal input. Included stability and parameter-sensitivity analysis, delivered through an interactive web interface.



04 Cancer Survival Prediction

Machine Learning · Python · Healthcare Data

ML pipeline predicting cancer survival from TCGA clinical data — preprocessing, encoding, feature scaling, multiple model comparison, K-fold cross-validation and bootstrapping, evaluated via accuracy and ROC/AUC.

EXPERIENCE

Advanced Research Intern — NDI (Northern Digital Inc.)

Engineering & research on biomedical tracking technologies within NDI's Advanced Research team.

May – Aug 2026
Waterloo, ON

Project Management IT Intern — Oak Valley Health

Supported MEDITECH Expanse go-live (CPOE, Web Ambulatory); at-the-elbow clinical support; explored AI/ML for ticket classification.

Sep – Dec 2025

Engineering Intern — TanenbaumCHAT

Hardware/software prototyping with Python, MATLAB, Java, Arduino, and 3D printing.

Jan – Apr 2025

Python & GenAI Instructor — CodingPlusOne

Taught Python programming remotely to students ages 6–16, introducing generative AI concepts.

May – Aug 2024

OTHER PROJECTS & WORK

TURBO Walker — Anti-tilt assistive walker w/ automated braking; QFD-driven design.

Pokémon Battle Arena — Arduino + C++ real-time 2-player game using Timer1 interrupts & Java Processing UI.

A-Mode Ultrasound — C++ beamforming for ultrasound signal processing.

BioTron — Team lead on a dual-sensor wearable for post-lumbar-surgery posture monitoring.

Butterfly Puzzle — Mechanical toy with a wing-flapping mechanism, iterative design.

Hospital Volunteer — 150+ hrs, Orillia Soldiers' Memorial Hospital (porter & front desk).